Swaroop Kumar Yadav

Bachelor of Technology
Electronics and Communication Engineering
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Education

- **Bachelor of Technology in Electronics and Communication Engineering** *2022 – 2026*
  Hemwati Nandan Bahuguna Garhwal University, Srinagar, Uttarakhand

Technical Skills and Interests

Languages: C/C++, Python, VHDL, Verilog
Electronics Design Tools: Xilinx Vivado, Altera Quartus, LTspice, Proteus, Electric Binary
Areas of Interest: VLSI, RTL Design, Digital ASIC Flow (Conceptual), Embedded Systems Design, IoT Applications.

Experience

- **Summer Internship – Telecom Regulatory Authority of India (TRAI)** *June 2025 – August 2025*
  NSL-I Division (Network Spectrum & Licensing) New Delhi, India
  - Contributed technical analysis to TRAI Consultation Paper No. 11/2025 on interconnection regulation.
  - Worked on mobile number porting and satellite–5G integration studies with references to 3GPP and NTN standards.

Publications

- **Adaptive Traffic Signal Control System Based on Ultrasonic Sensors using Kalman Filter** *IC2E3, 2025*
  2025 IEEE International Conference on Computer, Electronics, Electrical Engineering & their Applications (IC2E3)
  - Published a peer-reviewed IEEE conference paper proposing an adaptive traffic signal control system using ultrasonic sensing and Kalman filtering for real-time traffic optimization.
  - DOI: 10.1109/IC2E365635.2025.11167485

Projects

- **8-bit Soft-Core CPU Design based on SAP-1 Architecture (VHDL)** *2025–26*
  Designed and implemented a complete 8-bit soft-core CPU based on the SAP-1 (Von Neumann) architecture.
  - Implemented a single-cycle, accumulator-based CPU with a shared 8-bit tri-state data bus.
  - Designed a hardwired Control Unit using FSM-based T-states (T0–T6) to control fetch–decode–execute sequencing.
  - Implemented ALU (ADD, SUB), Program Counter, MAR, Instruction Register, registers, output register, and 16×8 RAM.
  - Defined and validated a 7-instruction ISA: LDA, STA, ADD, SUB, JMP, OUT, HLT.

- **VHDL-Based Digital Logic Module Library** *April 2025*
  Collection of reusable combinational and sequential digital building blocks.
  - Implemented and verified a modular 24-hour digital clock architecture.
  - Implemented display and sensor-interfacing logic for 7-segment, 16×2 LCD, VGA output, and basic sensor modules.
  - Designed adders, multiplexers, encoders, decoders, counters, shift registers, multipliers, and FSM-based logic.
  - GitHub Link: https://s2sofficial.github.io/vhdl

- **Bare-Metal Programming on 8051 Microcontroller** *Feb 2025*
  Register-level peripheral control using Assembly language on classic 8-bit microcontrollers.
  - Implemented LCD, ADC, RTC, timers, counters, UART, and SPI interfacing using pure 8051 Assembly language.
  - Developed timing-accurate delay routines and peripheral drivers through direct register manipulation.

- **Embedded Systems Projects (Arduino, ESP32, ESP8266)** *2024-25*
  Sensor-based embedded applications involving real-time monitoring and control.
  - Developed an adaptive traffic control system using ultrasonic sensors and real-time traffic density estimation, forming the experimental basis for a published IEEE conference paper.
  - Designed a solar panel sun-tracking system using Arduino UNO to maximize energy harvesting.
  - Implemented a Wi-Fi analyzer on ESP32 that measures RSSI and network latency.
  - Built an RFID-based secure access control system with unique tag identification and authorization logic.
  - Developed a water quality monitoring system using turbidity sensors.